

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK 2

MARCH 2004

MATHEMATICS 2 UNIT

Time allowed: 70 minutes

Instructions:

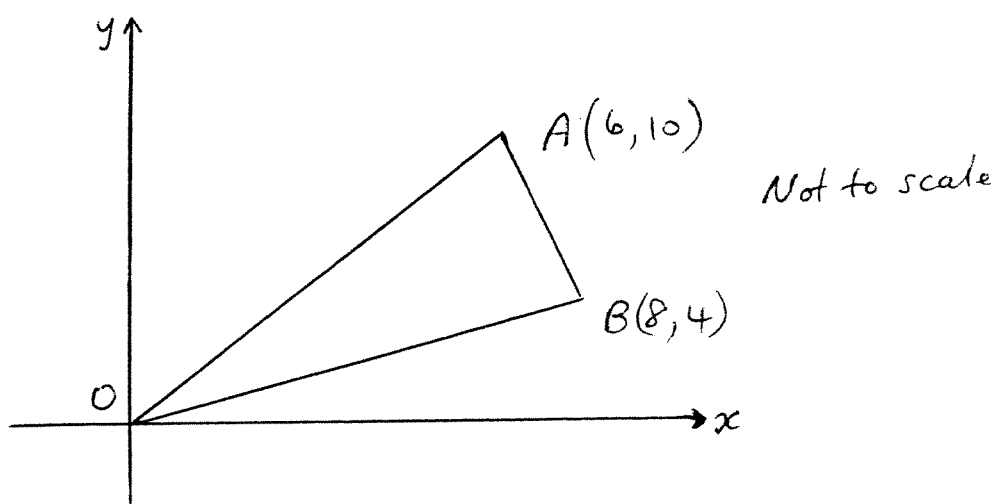
- Start each question on a new page
- Full marks may not be awarded if working is incomplete or illegible.

Name: _____

Teacher: _____

Q1	Q2	Q3	Q4	Q5	Total
/10	/10	/10	/10	/10	/50

Question 1



- Copy this diagram above onto your answer page
- Find the length of OA in surd form.
- Find the gradient of OA

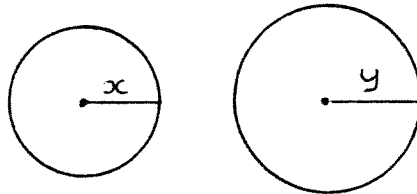
1

1

- d) Find the equation of OA in general form 2
- e) K is on OA such that $BK \perp OA$. Use your answer in part (c) to find the gradient of BK and hence the equation of BK. 2
- f) Find the perpendicular distance BK and hence the area of $\triangle OAB$. 3
- g) Find the coordinates of a point C such that OABC is a parallelogram 1

Question 2

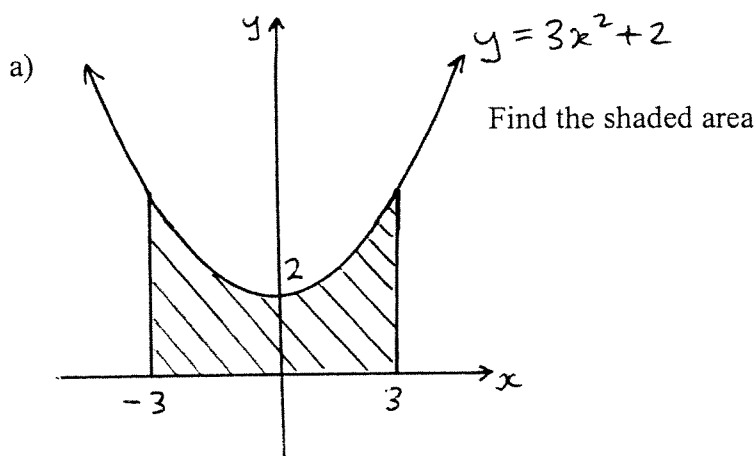
- a) Given $\frac{dy}{dt} = 12t^2 + t$ and that $y = 20$ when $t = 2$, find y in terms of t 2
- b) (i) Find $\int (x^2 + \frac{3}{x^2}) dx$ 2
- (ii) Evaluate $\int_1^3 \frac{1}{\sqrt{x}} dx$ 2
- c) Two circles are such that the sum of their radii is constant at 10 cm.



Let the radii of the two circles be x cm and y cm.

- (i) Show that the sum of their areas is given by $A = 2\pi x^2 - 20\pi x + 100\pi$ 1
- (ii) Show that the sum of their areas will be a minimum when the radii are equal 3

Question 3



- b) (i) Find the x values of the points of intersection of the graphs of $y = x^2 - 6x$ and $y = 2x$. 1
- (ii) Find the area between the two graphs above. 3
- c) Find the area between the curve $y = x^3$, the y -axis and the lines $y = 1$ and $y = 8$ 3

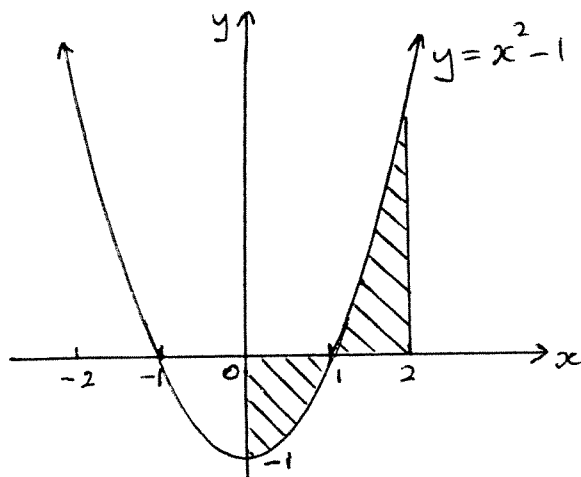
Question 4

- a) Find a primitive of $(3x + 2)^4$ 1
- b) For the curve $y = 6x^2 - x^3 + 9$:
- (i) Find the stationary points and determine their nature 4
 - (ii) Determine any point (s) of inflexion 2
 - (iii) Sketch the curve over the domain $-3 \leq x \leq 5$ 2
 - (iv) In this domain, determine the maximum value of the function. 1

Question 5

- a) Sketch a curve on a number plane that satisfies all conditions below: 3
- $f(0) = 2$
- When $x < 0$, $f'(x) < 0$ and $f''(x) > 0$
- When $x > 0$, $f'(x) < 0$ and $f''(x) < 0$.

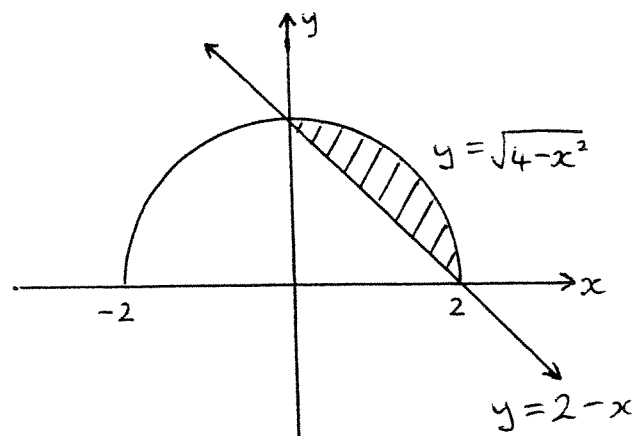
- b) 3



Find the shaded area

- c) The graphs of $y = 2 - x$ and $y = \sqrt{4 - x^2}$ are shown. 4

Find the volume generated when
the shaded area shown
is rotated about the x -axis.



Solutions.

$$\textcircled{1} \text{ b) } OA = \sqrt{6^2 + 10^2} \\ = \sqrt{136} \quad \textcircled{1}$$

$$\text{c) } m_{AB} = \frac{10}{6} \\ = \frac{5}{3} \quad \textcircled{1}$$

$$\text{d) } y = \frac{5}{3}x, \quad 3y = 5x \\ \textcircled{1} \quad \therefore 5x - 3y = 0 \quad \textcircled{1}$$

$$\text{e) } m_{BK} = -\frac{3}{5} \quad \textcircled{1}$$

$\therefore$ eqn. of BK is

$$y - 4 = -\frac{3}{5}(x - 8)$$

$$5y - 20 = -3x + 24$$

$$\therefore 3x + 5y - 44 = 0 \quad \textcircled{1}$$

f) line $5x - 3y = 0$
point $(8, 4)$

$$\therefore p.d. = \frac{5 \times 8 + (-3) \times 4 + 0}{\sqrt{5^2 + 3^2}}$$

$$= \frac{28}{\sqrt{34}} \quad \textcircled{1} \text{ correct idea}$$

$$\frac{28}{\sqrt{34}} \quad \textcircled{1} \text{ correct applic.}$$

$$\therefore \text{area } \triangle OAB = \frac{1}{2} \times \sqrt{136} \times \frac{28}{\sqrt{34}} \\ = 28u^2 \quad \textcircled{1}$$

$$\text{g) } C(2, -6) \quad \textcircled{1}$$

$$\textcircled{2} \text{ a) } y = 4t^3 + \frac{t^2}{2} + c \quad \textcircled{1}$$

$$y = 20, t = 2 :$$

$$\therefore 20 = 32 + 2 + c \quad (c = -14)$$

$$\therefore y = 4t^3 + \frac{t^2}{2} - 14 \quad \textcircled{1}$$

$$\text{b) (i) } \int (x^2 + 3x^{-2}) dx \\ = \frac{x^3}{3} + 3 \frac{x^{-1}}{-1} + c \quad \textcircled{1} \\ = \frac{x^3}{3} - \frac{3}{x} + c \quad \textcircled{1}$$

$$\text{(ii) } \int_1^3 x^{-1/2} dx = \left[\frac{x^{1/2}}{1/2} \right]_1^3 \quad \textcircled{1} \\ = [2\sqrt{x}]_1^3 \\ = 2\sqrt{3} - 2 \quad \textcircled{1}$$

$$\text{c) (i) } A = \pi x^2 + \pi y^2 \\ \text{(and } x + y = 10 \\ \therefore y = 10 - x)$$

$$\therefore A = \pi x^2 + \pi (10 - x)^2 \\ = \pi x^2 + \pi (100 - 20x + x^2) \quad \textcircled{1} \\ = \pi x^2 + 100\pi - 20\pi x + \pi x^2 \\ = 2\pi x^2 - 20\pi x + 100\pi$$

(ii) min. area when $\frac{dA}{dx} = 0$ ①

$$\frac{dA}{dx} = 4\pi x - 20\pi = 0$$

$$4\pi(x-5) = 0$$

$$\therefore x = 5 \quad ①$$

x	5^-	5	5^+	
$\frac{dA}{dx}$	-	0	+	①

$\therefore$ area must be a minimum when $x = 5$ and $y = 5$, i.e. equal radii.

③ a) $A = 2 \times \int_0^3 (3x^2 + 2) dx$ ①

$$= 2 [x^3 + 2x]_0^3 \quad ①$$

$$= 2 [27 + 6 - (0 + 0)]$$

$$= 66 u^2 \quad ①$$

b) (i) At intersection,

$$x^2 - 6x = 2x$$

$$x^2 - 8x = 0$$

$$x(x-8) = 0$$

$$\therefore x = 0 \text{ or } 8 \quad ①$$

(ii) $A = \left| \int_0^8 (x^2 - 6x - 2x) dx \right|$ ①

$$= \left| \int_0^8 (x^2 - 8x) dx \right|$$

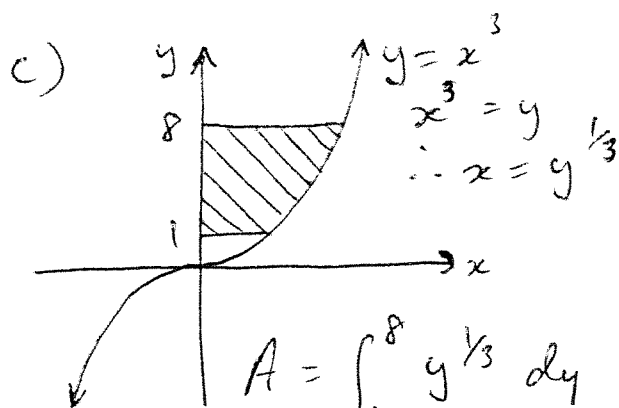
$$= \left| \left[\frac{x^3}{3} - 4x^2 \right]_0^8 \right| \quad ①$$

$$= \left| \left(\frac{512}{3} - 256 \right) - (0 - 0) \right|$$

$$= \left| 170\frac{2}{3} - 256 \right|$$

$$= \left| -85\frac{1}{3} \right|$$

$$= 85\frac{1}{3} u^2 \quad ①$$



$$A = \int_1^8 y^{1/3} dy \quad ①$$

$$= \left[\frac{3}{4} y^{4/3} \right]_1^8 \quad ①$$

$$= \frac{3}{4} (8^{4/3} - 1^{4/3})$$

$$= \frac{3}{4} (16 - 1)$$

$$= \frac{3}{4} \times 15$$

$$= 11\frac{1}{4} u^2 \quad ①$$

④ a) $\frac{(3x+2)^5}{5 \times 3} + c$

$$= \frac{(3x+2)^5}{15} + c \quad ①$$

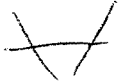
b) (i) S.P.'s when $\frac{dy}{dx} = 0$ ①

$$\frac{dy}{dx} = 12x - 3x^2 = 0$$

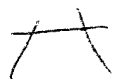
$$3x(4-x) = 0$$

$$\therefore x = 0, 4. \quad ①$$

x	0^-	0	0^+
$\frac{dy}{dx}$	$-$	0	$+$



x	4^-	4	4^+
$\frac{dy}{dx}$	$+$	0	$-$



$\therefore$ min. T.P. at $(0, 9)$ ①

$\therefore$ max. T.P. at $(4, 41)$ ①

ii) P. of I. when $\frac{d^2y}{dx^2} = 0$
and concavity changes

$$\frac{d^2y}{dx^2} = 12 - 6x = 0$$

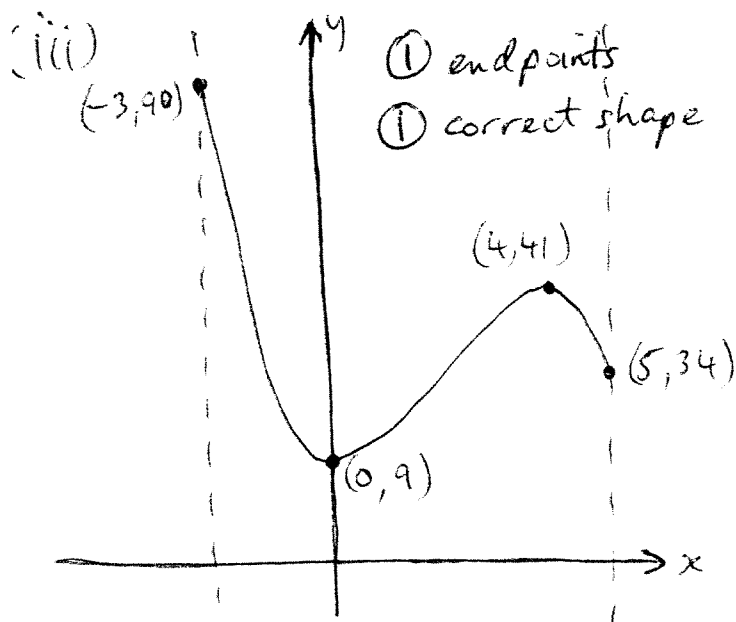
$$\therefore x = 2 \quad ①$$

x	2^-	2	2^+
$\frac{d^2y}{dx^2}$	$+$	0	$-$

①

$\therefore$ concavity changes

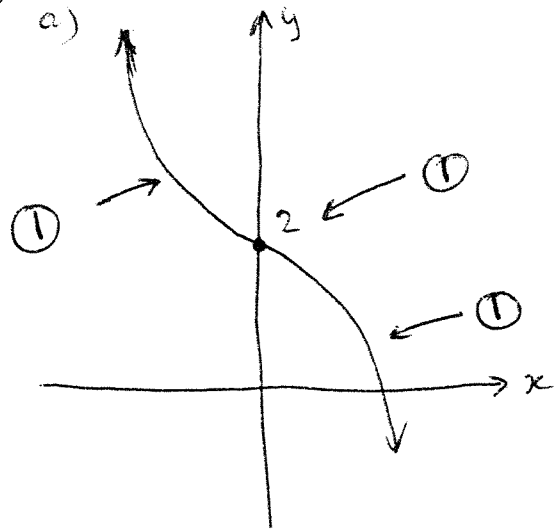
$\therefore$ P. of I. at $(2, 25)$



(iv) maximum value is 90
or $y = 90$ ①

⑤

a)



b) $A = \left| \int_0^1 (x^2 - 1) dx \right| + \int_1^2 (x^2 - 1) dx$ ①

$$= \left| \left[\frac{x^3}{3} - x \right]_0^1 \right| + \left[\frac{x^3}{3} - x \right]_1^2$$

$$= \left| \left(\frac{1}{3} - 1 \right) - (0 - 0) \right| + \left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right)$$

$$= \left| -\frac{2}{3} \right| + \frac{2}{3} - \left(-\frac{2}{3} \right)$$

$$= \frac{2}{3} + \frac{2}{3} + \frac{2}{3}$$

$$= 2 \quad ①$$

c) $V = \pi \int_0^2 (\sqrt{4-x^2})^2 - (2-x)^2 dx$ ①

$$= \pi \int_0^2 4 - x^2 - (4 - 4x + x^2) dx$$

$$= \pi \int_0^2 (4 - x^2 - 4 + 4x - x^2) dx$$

$$= \pi \int_0^2 (4x - 2x^2) dx \quad ①$$

$$= \pi \left[2x^2 - \frac{2x^3}{3} \right]_0^2 \quad (1)$$

$$= \pi \left[\left(8 - \frac{16}{3} \right) - (0 - 0) \right]$$

$$= \pi \times 2^{\frac{2}{3}}$$

$$= \frac{8\pi}{3} u^3 \quad (1)$$